

TEST YOUR UNDERSTANDING

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Levels 1-2: Remembering/Understanding

1. Binding of a signaling molecule to which type of receptor leads directly to a change in the distribution of substances on opposite sides of the membrane?
(A) intracellular receptor
(B) G protein-coupled receptor
(C) phosphorylated receptor tyrosine kinase dimer
(D) ligand-gated ion channel
2. The activation of receptor tyrosine kinases is characterized by
(A) dimerization and phosphorylation.
(B) dimerization and IP₃ binding.
(C) a phosphorylation cascade.
(D) GTP hydrolysis.
3. Lipid-soluble signaling molecules, such as aldosterone, cross the membranes of all cells but affect only target cells because
(A) only target cells retain the appropriate DNA segments.
(B) intracellular receptors are present only in target cells.
(C) only target cells have enzymes that break down aldosterone.
(D) only in target cells is aldosterone able to initiate the phosphorylation cascade that turns genes on.
4. Consider this pathway: epinephrine → G protein-coupled receptor → G protein → adenylyl cyclase → cAMP. Identify the second messenger.
(A) cAMP
(B) G protein
(C) GTP
(D) adenylyl cyclase
5. Which of the following occurs during apoptosis?
(A) lysis of the cell
(B) direct contact between signaling cells
(C) fragmentation of the DNA
(D) release of proteases outside the cell

Levels 3-4: Applying/Analyzing

6. Which observation suggested to Sutherland the involvement of a second messenger in epinephrine's effect on liver cells?
(A) Enzymatic activity was proportional to the amount of calcium added to a cell-free extract.
(B) Receptor studies indicated that epinephrine was a ligand.
(C) Glycogen breakdown was observed only when epinephrine was administered to intact cells.
(D) Glycogen breakdown was observed only when epinephrine and glycogen phosphorylase were mixed.
7. Protein phosphorylation is commonly involved with which of the following?
(A) ligand binding by receptor tyrosine kinases.
(B) activation of G protein-coupled receptors.
(C) activation of protein kinase molecules.
(D) release of Ca²⁺ from the ER lumen.

Levels 5-6: Evaluating/Creating

8. **DRAW IT** Draw the following apoptotic pathway, which operates in human immune cells. A death signal is received when a molecule called Fas binds its cell-surface receptor. The binding of many Fas molecules to receptors causes receptor clustering. The intracellular regions of the receptors, when together, bind proteins called adaptor proteins. These in turn bind to inactive molecules of caspase-8, which become activated and then activate caspase-3. Once activated, caspase-3 initiates apoptosis.

9. **EVOLUTION CONNECTION** Identify the evolutionary mechanisms that might account for the origin and persistence of cell-to-cell signaling systems in prokaryotes.

10. **SCIENTIFIC INQUIRY** Epinephrine initiates a signal transduction pathway that produces cyclic AMP (cAMP) and leads to the breakdown of glycogen to glucose, a major energy source for cells. But glycogen breakdown is only part of the fight-or-flight response that epinephrine brings about; the overall effect on the body includes an increase in heart rate and alertness, as well as a burst of energy. Given that caffeine blocks the activity of cAMP phosphodiesterase, propose a mechanism by which caffeine ingestion leads to heightened alertness and sleeplessness.

11. **SCIENCE, TECHNOLOGY, AND SOCIETY** The aging process is thought to be initiated at the cellular level. Among the changes that can occur after a certain number of cell divisions is the loss of a cell's ability to respond to growth factors and other signals. Much research into aging is aimed at understanding such losses, with the ultimate goal of extending the human life span. Not everyone, however, agrees that this is a desirable goal. If life expectancy were greatly increased, discuss what might be the social and ecological consequences.

12. **WRITE ABOUT A THEME: ORGANIZATION** The properties of life emerge at the biological level of the cell. The highly regulated process of apoptosis is not simply the destruction of a cell; it is also an emergent property. Write a short essay (about 100–150 words) that briefly explains the role of apoptosis in the development and proper functioning of an animal, and describe how this form of programmed cell death is a process that emerges from the orderly integration of signaling pathways.

13. SYNTHESIZE YOUR KNOWLEDGE



There are five basic tastes—sour, salty, sweet, bitter, and “umami.” Salt is detected when the concentration of salt outside of a taste bud cell is higher than that inside of it, and ion channels allow the passive leakage of Na⁺ into the cell. The resulting change in membrane potential (see Concept 7.4) sends the “salty” signal to the brain. Umami is a savory taste generated by glutamate (glutamic acid, found in monosodium glutamate, or MSG), which is used as a flavor enhancer in foods such as taco-flavored tortilla chips. The glutamate receptor is a GPCR, which, when bound, initiates a signaling pathway that ends with a cellular response, perceived by you as “taste.” If you eat a regular potato chip and then rinse your mouth, you will no longer taste salt. But if you eat a flavored tortilla chip and then rinse, the taste persists. (Try it!) Propose a possible explanation for this difference.

For selected answers, see Appendix A.